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Toward an Improved Auditory Model for Predicting Binaural Coloration

THOMAS MCKENZIE,[1] *AES Member* , **AND FABIAN BRINKMANN,**[2] *AES Fellow*
(thomas.mckenzie@ed.ac.uk) (fabian.brinkmann@tu-berlin.de)

[1]*Acoustics and Audio Group, Reid School of Music, University of Edinburgh, Edinburgh, UK*
[2]*Audio Communication Group, Technische Universität Berlin, Berlin, Germany*

The evaluation of audio quality is important in the development of immersive audio algorithms and reproduction systems, and binaural models are often used for this as a quick alternative to listening tests. Coloration (i.e., perceived loudness differences integrated across ears and frequency) is one key quality aspect; however, the majority of models used to predict coloration are often oversimplified or are missing a dedicated binaural stage to consider the relative contribution of the left and right ear signals. A binaural coloration model is presented that builds upon previous work and tests three different approaches for its binaural stage. The proposed model is evaluated in comparison with nine models that are frequently used to predict coloration by using data from five listening tests totaling 252 stimuli with various audio contents and source positions. The proposed model performed best with 85% of explained variance, followed by predictions based on ISO 532-1 loudness, yielding 78% explained variance. The commonly used log-spectral distance performed worst, with only 44% explained variance. The three tested binaural stages had little influence on the performance of the proposed model. The model is made freely available to download.

0 INTRODUCTION

The evaluation of audio quality is important for assessing the degree of realism with which immersive audio algorithms (e.g., binaural synthesis or beamforming) and reproduction systems (e.g., Ambisonics, wave-field synthesis, or channel-based surround sound) can reproduce acoustic environments. While initial research on the quality of immersive audio systems historically focused on localization accuracy (e.g. [1, 2]), a variety of specifically tailored quality measures emerged in the following decades [3]. One category of measures are *quality inventories* that contain a list of perceptual qualities against which spatial audio systems can be evaluated [4, 5]. Among these qualities, coloration is consistently seen as one of the most important aspects with respect to the overall quality, dominating the perception of basic audio quality more so than localization [6].

Because coloration is a perceptual quality, the most valid way to measure it is through listening tests. The multiple stimulus test with hidden reference and anchor (MUSHRA [7]) allows for direct comparison of several test conditions, and is often adapted to asses a variety of audio quality aspects, including coloration. However, listening tests are time consuming and often not feasible during the development of immersive audio reproduction systems, where a plethora of design choices that may affect coloration need to be taken. For this purpose, alternative assessment strategies are desirable, and a common choice is to develop auditory models that aim at predicting specific auditory qualities. However, this requires a thorough understanding of coloration as a perceptual attribute.

Lindau et al. [4] and Wierstorf et al. [8] describe coloration as a *change/difference in timbre* with respect to an explicitly represented or implicit internal reference. Simon et al. [5] define it as the *feeling of a sound being richer* and *(perceived) spectral modifications* in certain frequency ranges. These definitions of coloration seem somewhat vague. The ambiguity partly stems from the imprecise but common definition of timbre as enabling "a listener to judge that two non-identical sounds, similarly presented and having the same loudness, pitch, spatial location, and duration, are dissimilar." [9, p. 59]. McAdams [10], offers a more detailed definition of timbre as a complex and multidimensional auditory attribute encompassing spectral, temporal, and spectrotemporal qualities such as brightness, roughness, attack quality, and inharmonicity among others, which is now also refelcted in the latest version of [9].

*To whom correspondence should be addressed, email: thomas.mckenzie@ed.ac.uk

Evaluation of binaural audio reproduction systems is often performed on the binaural signals. In fact, even loudspeaker reproduction systems are often evaluated binaurally through the use of binaural measurements (either individual or using a dummy head). Evaluations are usually made against a reference system, and identical audio signals (e.g., anechoic speech, music, or noise) are used as input for the reference system and system under test. In many such cases, temporal aspects of timbre may be disregarded to arrive at a simplified definition considering only spectral aspects. Coloration is thus defined in this paper as *binaural loudness differences between a reference and test stimulus that are integrated across ears and frequency by the auditory system*, where loudness is defined as the perceived intensity of a sound [11, Chapter 11]. While this definition is similar to the understanding of Simon et al. [5], it explicitly builds upon the related concepts of timbre and loudness, and emphasizes the need for auditory integration, two aspects that require attention when predicting/modeling coloration.

Note that disregarding temporal aspects might be considered an oversimplification in some cases (e.g., for loudspeaker-based reproduction), where contributions from multiple speakers and reflection from the room might affect temporal aspects of timbre and coloration. However, it might also be argued that only changes in the magnitude spectrum due to such effects relate to coloration, whereas temporal changes of the signal manifest themselves in spatial qualities such as the perception of source width, reverberation, and echos. In addition, it is debatable how overall loudness differences between stimuli should be considered. It might be intuitive to explicitly exclude them from the definition above; however, subjects might still report coloration for stimuli that are identical apart from a scalar gain factor even if being instructed to ignore overall loudness differences. It might therefore be most practical to keep the definition more general and account for overall loudness differences during test design if desired.

Despite the importance of coloration, there are very few specifically tailored auditory models for its prediction. As a consequence, very simple models for spectral differences are often used to predict coloration. The Auditory Modeling Toolbox (AMT) may be the largest collection of auditory models [12], containing more than 15 models for spatial perception and four loudness models, but it contains only a single model that aims at predicting coloration between two stimuli, which is the model from McKenzie et al. [13]. A second coloration model, based on the so called composite loudness level (CLL) [14], has been used to predict coloration for immersive audio reproduction systems based on amplitude or time-panning [14–16]. A third coloration model developed by Moore and Tan [17] and implemented in the TWO!EARS project [18] has been used to predict the coloration induced by wave field synthesis [19].

Additionally, the AMT contains models that predict the overall audio quality [20–22] using peripheral ear models followed by feature extraction stages. The features are then mapped to a perceptual quality scale based on experimental data. More integrative models exist outside the AMT, such as the generative machine listener [23] or the joined monaural and binaural quality model [24], which both predict the overall audio quality, and a model for predicting just noticeable differences between two head-related transfer functions (HRTFs) [25]. However, these models do not distinguish between coloration and other qualities, such as localization or reverberation, which might be desirable when developing systems for immersive audio reproduction.

The current study aims to (i) provide a comprehensive comparison of existing models that are frequently used to predict binaural coloration, (ii) develop an improved coloration auditory model, and (iii) test the influence of different binaural stages for computing the scalar value for predicted coloration. The proposed model is inspired by a previously developed model [13] that performed best in a comparison with three other models. The comparison uses an extended database of stimuli and perceptual ratings and an increased number of existing models for benchmarking.

The paper is organized as follows: SEC. 1 introduces existing models for predicting binaural coloration. SEC. 2 presents the methodology of the binaural coloration auditory model. SEC. 3 then presents an evaluation of the proposed model against the existing models by comparing their predictions to coloration ratings from five listening tests. The results of the evaluation are discussed in SEC. 4, identifying the models that perform the strongest, suggesting why, and discussing limitations of the evaluation. Finally, the paper is concluded in SEC. 5 along with future work to further improve the model.

1 BACKGROUND

In this paper, $h_{\text{ref}}[n]$ and $h_{\text{test}}[n]$ are used to denote reference and test input signals, respectively, with $1 \leq n \leq N$ denoting the sample number up to a signal duration of N samples. The input signals in the frequency domain are calculated using the fast Fourier transform (FFT) to give $H[k]$ with $1 \leq k \leq K$ denoting the index of each frequency bin, with the total number of frequency bins $K = N$. The sample rate is denoted by f_s. A binaural signal is written as $h^{\text{l,r}}[n]$, with superscript l and r denoting left and right, respectively. Coloration models described in this study often apply identical processing for the left and right ear signals, in which case the superscripts l and r are omitted. If not explicitly specified, the final scalar value for coloration is obtained by taking the arithmetic mean across frequencies and ears.

Due to the lack of dedicated coloration models, many previous studies employed simple models as linear predictors for coloration. The simplest way to numerically estimate the coloration between audio signals is to calculate the difference between the magnitude spectra of the signals. The basic spectral difference (BSD) computes the mean absolute difference in decibels between a reference spectrum H_{ref} and test spectrum H_{test} as

$$\text{BSD} = \frac{1}{K} \sum_{k=1}^{K} \left| 20 \log_{10} \frac{|H_{\text{test}}[k]|}{|H_{\text{ref}}[k]|} \right| . \tag{1}$$

Similarly, the log spectral distance (LSD, also called log spectral distortion) performs a root mean square average as

$$\mathrm{LSD} = \sqrt{\frac{1}{K}\sum_{k=1}^{K}\left(20\log_{10}\frac{|H_{\mathrm{test}}[k]|}{|H_{\mathrm{ref}}[k]|}\right)^2}. \tag{2}$$

BSD and LSD operate on level differences in decibels of each frequency bin as a simple approximation for loudness differences. However, they are assumed to be biased toward high-frequency differences due to the linear frequency resolution of the FFT as opposed to the logarithmic frequency resolution of the auditory system. To address this, the energy of each signal within $K_A = 40$ auditory filters $W[k, f_c]$ with center frequencies f_c distributed on an equivalent rectangular bandwidth (ERB) frequency scale [26] between 20 Hz and 20 kHz can be calculated as

$$H[f_c] = \sum_{k=1}^{K} |W[k, f_c]|\,|H[k]|^2 , \tag{3}$$

and the auditory band BSD can be computed as

$$\mathrm{BSD_A} = \frac{1}{K_A}\sum_{f_c=1}^{K_A}\left|10\log_{10}\frac{H_{\mathrm{test}}[f_c]}{H_{\mathrm{ref}}[f_c]}\right| . \tag{4}$$

Building further on this, Moore and Tan introduced two metrics [17] that not only incorporate ERB frequency weighting but also attempt to approximate the frequency-dependent sensitivity of human hearing. These use the weighted sum of first and second-order differences between ERB-spaced frequency bands. The two metrics are the standard deviation (herein labeled $\mathrm{Moore04_{SD}}$) and sum (herein labeled $\mathrm{Moore04_{sum}}$) of differences. An additional frequency-dependent weighting function is also included that progressively reduces the amplitude of frequencies above approximately 1 kHz (with additional weighting for speech signals, removing frequencies below approximately 125 Hz and above 11 kHz). This is a first step toward more detailed implementation, such as equal loudness contours, as given in International Organization for Standardization (ISO) 226 [27].

CLL is another simple model that uses half-wave rectification and 1-kHz low-pass filtering to simulate the hair cell and auditory nerve behavior and a gammatone filterbank and equal loudness frequency weighting [14]. Some implementations of CLL use the adaptation network of [28], which is a multilayer feedforward neural network to acquire mappings and properties from the front end. The CLL has frequently been used to predict coloration but is missing a dedicated binaural stage.

Zwicker [29, 30] developed models for approximating the summation of loudness across frequency that make up ISO 532-1 [31]. Here, the Bark frequency scale is used [32] (which is conceptually similar to the ERB frequency scale). These models have since been extended by Moore and Glasberg [33, 34] with further considerations, including a binaural weighting referred to as a binaural inhibition stage [35], and make up ISO 532-2 [36]. Note that this makes ISO 532-2 the only established model with a dedicated binaural stage as opposed to a mean or sum of the left and right coloration values.

The previously published Predicted Binaural Coloration (PBC-1 [13]) model uses equal loudness weighting through ISO 226 Phon curves [27], loudness weighting using the sone scale, and ERB frequency weighting. Note that the ERB weighting employed in PBC-1 differs from other models (Moore04, $\mathrm{BSD_A}$, CLL, and ISO 532) in that, instead of averaging over ERB frequency bands, it retains the number of frequency bins but weights them accordingly.

To the authors' best knowledge, the PBC-1 study is the only systematic evaluation of coloration methods against perceived coloration ratings from a listening test using binaural signals. Compared with BSD, CLL, and the more integrative Perceptual Audio Quality (PEAQ) [20] measure, the PBC-1 model showed the highest correlation with the listening test ratings. Nevertheless, this is a simple model, and a binaural PBC-1 value is a summation of the left and right loudness values.

1.1 Toward an Improved Prediction

Using a larger pool of binaural stimuli and coloration ratings (as detailed in SEC. 3.1), a number of possible improvements to the PBC-1 methodology were identified. The use of a sone scale for loudness weighting of frequency bins significantly affected the comparability of coloration values between listening tests: values for stimuli that produced similar ranges of perceptual ratings could differ by as much as a factor of 10 between tests. The model also lacked signal rectification, high-frequency spectral smoothing, and a dedicated binaural weighting stage.

This paper attempts to address the shortcomings of the PBC-1 and other published loudness models often used to predict coloration, presenting an updated model that is herein referred to PBC-2. The aim is to produce a more accurate prediction of binaural coloration perception and one that produces results that are better comparable between different tests and stimuli groups.

2 METHOD

A block diagram of the proposed method is presented in Fig. 1. The PBC-2 method builds on the PBC-1 method: additional input signal preprocessing is implemented, the human auditory system is modeled in more detail, and the binaural weighting incorporates interaural time and level differences in the calculations. The sone loudness weighting is removed.

The model asserts that input signals have two channels (binaural) and that the input reference and test signal sets have the same number of signals. The tails of the two input signal sets are zero padded to ensure they match in length. This allows for comparisons with dry and reverberant signals, for example, or impulse responses with different durations, such as head-related impulse responses (HRIRs) with different filter lengths.

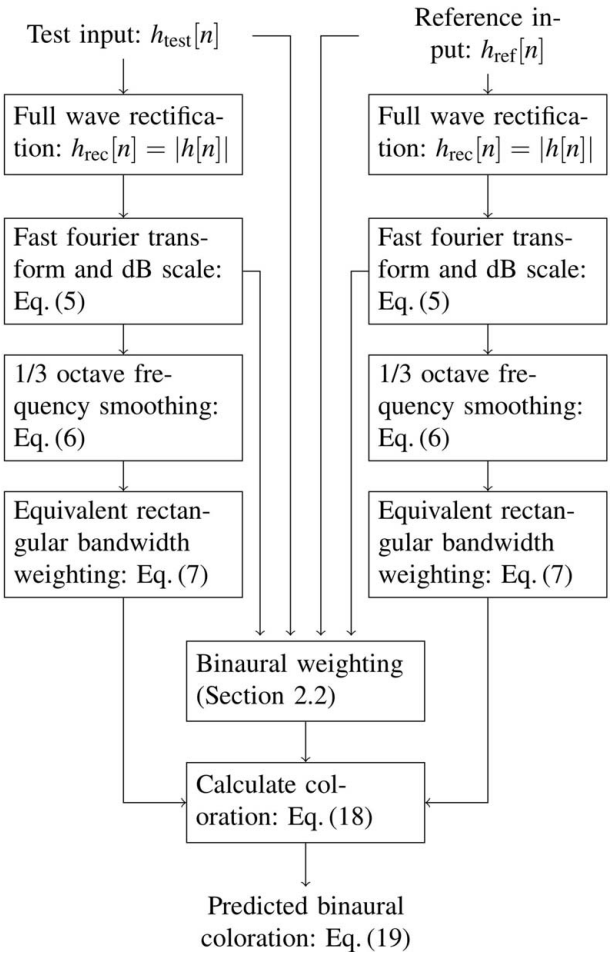

Fig. 1. Block diagram summary of PBC-2.

2.1 Peripheral Model

Before the conversion of the signals from the time domain to the frequency domain, processing to model the behavior of the inner ear is performed. The first is the use of full-wave rectification: $h_{\text{rec}}[n] = |h[n]|$, which approximates some of the behavior of the hair cells [37]. While half-wave rectification is often used in perceptual models, full-wave rectification is preferred in this model to retain more signal information because phase of the playback system is unknown.

The time domain signals h_{rec} are then converted to the frequency domain, and the amplitude values of the frequency bins are converted to decibels as

$$L[k] = 10 \log_{10} |\mathfrak{F}(h_{\text{rec}}[n])|^2 , \tag{5}$$

where $L[k]$ can be understood as a simple estimate of the loudness in decibels and $\mathfrak{F}$ denotes the FFT operation. The frequency domain signals are now truncated to the specified frequency range of analysis, which by default is 20 Hz to 20 kHz.

While other models, such as CLL and ISO 532 use filterbanks for ERB weighting and frequency smoothing, this model retains individual frequency bins to preserve low-frequency spectral resolution. Third-octave smoothing is implemented above 5 kHz, where spectral details of HRTFs have been shown to be inaudible [38]. The smoothing of the frequency responses is achieved by averaging $L[k]$ within one-sixth octave frequency bands, and smoothed loudness values for every frequency bin $L_{\text{sm}}[k]$ are then calculated from the averaged values using Akima cubic Hermite interpolation [39] with MATLAB's *interp1* function. The transition from no smoothing to one-third octave smoothing is achieved as

$$L[k] = \begin{cases} (1 - \frac{k}{K_o})L[k] + \frac{k}{K_o} L_{\text{sm}}[k] & \text{for } f[k] \leq 5000 , \\ L_{\text{sm}}[k] & \text{otherwise} , \end{cases} \tag{6}$$

where K_o is the frequency bin for which $f[K_o] = 5$ kHz.

The magnitude value of each frequency bin is weighted according to its ERB value, using weights $g_{\text{ERB}}[k]$ that are computed as

$$f_{\text{ERB}}[k] = (0.108 f[k] + 24.7)^{w_{\text{ERB}}} \ \textit{and} \tag{7}$$

$$g_{\text{ERB}}[k] = f_{\text{ERB}}[1]/f_{\text{ERB}}[k] , \tag{8}$$

where $f[k] = f_s/N(k-1)$ is the frequency in Hz and $f_{\text{ERB}}[k]$ is the ERB weighted frequency. Also, w_{ERB} is an additional weighting of the ERB curve with a default value $w_{\text{ERB}} = 1.35$, which further reduces the relative weight of high frequencies, similar to what is seen in the Moore04 and CLL models.

2.2 Binaural Weighting

Three methods for binaural weighting are implemented. The first is a simple mean of the left and right coloration values (as used with LSD, BSD, and ISO 532-1): $g^{\text{l}}[k] = g^{\text{r}}[k] = 0.5$, where $g^{\text{l}}[k]$ and $g^{\text{r}}[k]$ are frequency-dependent binaural weights for the left and right signals, respectively. The other two binaural weightings are described in the forthcoming section; results are reported for all three.

2.2.1 Interaural Method

In the interaural binaural-weighting method, the left and right coloration values for each frequency bin are weighted based on interaural differences [29] of the reference signals, following the duplex theory [40]. Frequencies below 1.5 kHz are weighted using interaural time difference (ITD), and frequencies above 1.5 kHz are weighted using interaural level difference (ILD). The weighting is such that signals arriving earlier and/or louder at one ear (typically the ipsilateral ear) will be weighted higher.

ITD is estimated from the maximum of the cross-correlation, a method considered to produce accurate estimations [41]:

$$\text{ITD} = \frac{\arg\max_m \left(\sum_{n=1}^{N} h^{\text{l}}[n] \cdot h^{\text{r}}[n-m] \right)}{f_s} . \tag{9}$$

Note that this calculation is performed on the time domain signals, after the input signal preprocessing but before the peripheral model. ITD values are bound within the range of ± 0.8 ms, which is the approximate limit for a human head [42, 43].

ILD is known to be frequency dependent, where at maximum lateralization (90° azimuth), ILD is approximately 10 dB at 500 Hz and up to 30 dB at 5 kHz [44]. ILD is calcu-

lated above 1.5 kHz in the frequency domain separately for each frequency bin:

$$\mathrm{ILD}[k] = \frac{L^{\mathrm{l}}_{\mathrm{ref}}[k] - L^{\mathrm{r}}_{\mathrm{ref}}[k] + L^{\mathrm{l}}_{\mathrm{test}}[k] - L^{\mathrm{r}}_{\mathrm{test}}[k]}{2} . \quad (10)$$

The ILD values are then translated into weighting values. Based on preliminary testing, the sensitivity was chosen such that for an ILD of 10 dB, the ipsilateral weighting is approximately four times that of the contralateral:

$$g_{\mathrm{ILD}}[k] = 2^{(\mathrm{ILD}[k]/5)} . \quad (11)$$

Finally, the left and right ear gains are given by

$$g^{\mathrm{l}}[k] = \begin{cases} 0.5 + 125(\mathrm{ITD}_{\mathrm{ref}} + \mathrm{ITD}_{\mathrm{test}}) & \text{if } f[k] \leq 1500 , \\ g_{\mathrm{ILD}}[k]/(g_{\mathrm{ILD}}[k] + 1) & \text{otherwise} , \end{cases} \quad (12)$$

$$g^{\mathrm{r}}[k] = 1 - g^{\mathrm{l}}[k] . \quad (13)$$

The ITD sensitivity is such that for an ITD of 0.8 ms, the ipsilateral weighting is approximately two times that of the contralateral, in line with findings in [45].

2.2.2 Level Method

The interaural method described above assigns larger weights to the ipsilateral ear, which is similar to the binaural weighing applied in the the median plane localization model from Baumgartner et al. [46]. The idea behind this is that the louder ipsilateral ear signal is more important for the auditory impression. However, in the presence of coloration, the ipsilateral ear signal might not always be the loudest, and therefore the performance of a level-based averaging method was assessed to account for this.

In the first step, the maximum level is computed per ear and frequency as

$$L_{\max}[k] = \max(L_{\mathrm{ref}}[k], L_{\mathrm{test}}[k]) . \quad (14)$$

This is used compute the ILD based on the maximum level as

$$\mathrm{ILD}_{\max}[k] = L^{\mathrm{l}}_{\max}[k] - L^{\mathrm{r}}_{\max}[k] \quad (15)$$

and to derive the left and right ear gain as

$$g^{\mathrm{l}}[k] = \frac{2^{(\mathrm{ILD}_{\max}[k]/6)}}{1 + 2^{(\mathrm{ILD}_{\max}[k]/6)}} , \quad (16)$$

$$g^{\mathrm{r}}[k] = 1 - g^{\mathrm{l}}[k] . \quad (17)$$

The above will apply equal weights $g^{\mathrm{l}}[k] = g^{\mathrm{r}}[k] = 0.5$ if $\mathrm{ILD}_{\max}[k] = 0$. Based on preliminary testing, the sensitivity was chosen such that the weight of the louder ear doubles each 6 dB, that is $g^{\mathrm{l}}[k] = 2^n\, g^{\mathrm{r}}[k]$ for $\mathrm{ILD}_{\max}[k] = n\,6$ dB.

2.3 Final Coloration

The ERB weights as calculated in Eq. (7) are applied, as coloration between the test and reference signals is calculated as the absolute value of the subtraction of the test magnitudes from the reference magnitudes for each ear:

$$C[k] = \frac{g_{\mathrm{ERB}}[k]\, |L_{\mathrm{ref}}[k] - L_{\mathrm{test}}[k]|}{\sum_{k=1}^{K} g_{\mathrm{ERB}}[k]} . \quad (18)$$

Table 1. The 12 coloration methods used in the evaluation.

Model	Algorithm
BSD, Eq. (1)	Mean absolute difference
LSD, Eq. (2)	Root mean square difference
$\mathrm{BSD_A}$, Eq. (4)	Mean absolute difference in ERB-spaced frequency bands
$\mathrm{Moore04_{SD}}$ [17]	Standard deviation of first and second-order differences in ERB-spaced frequency bands
$\mathrm{Moore04_{sum}}$ [17]	Sum of first and second-order differences in ERB-spaced frequency bands
CLL [14]	Half-wave rectification, low-pass filtering, difference in gammatone filters
ISO 532-1 [31]	Loudness summation across frequency, Bark frequency scale
ISO 532-2 [36]	Loudness summation across frequency, Bark frequency scale and binaural inhibition stage
PBC-1 [13]	ERB frequency weighting, equal loudness weighting
$\mathrm{PBC\text{-}2_{mean}}$	Binaural weighting is mean of left and right coloration values
$\mathrm{PBC\text{-}2_{interaural}}$	Binaural weighting based on ITD ($f[k] \leq 1.5$ kHz) and level ($f[k] > 1.5$ kHz)
$\mathrm{PBC\text{-}2_{level}}$	Binaural weighting based on maximum level

Finally, the left and right coloration values are processed with the binaural weighting from one of the three methods in SEC. 2.2, and the results are summed to produce a single coloration value for each input binaural signal:

$$\mathrm{PBC\text{-}2} = \sum_{k=1}^{K} (g^{\mathrm{l}}[k]\, C^{\mathrm{l}}[k] + g^{\mathrm{r}}[k]\, C^{\mathrm{r}}[k]) . \quad (19)$$

3 EVALUATION

To evaluate the proposed PBC-2 model, perceived coloration ratings from previous listening tests were compared with predictions made using coloration methods and the associated binaural signals. Three versions (corresponding to the three binaural weighting methods) of PBC-2 were compared with nine other coloration methods, totaling 12 methods under evaluation as listed in Table 1 along with brief descriptions.

3.1 Test Data

To increase the robustness and validity of the evaluation, the pool of test signals was expanded from 56 [13] to 252 binaural signals featuring different audio contents and coloration types. This was achieved by pooling results from previous listening tests detailed in Table 2.

All listening tests followed a MUSHRA-like paradigm, where the similarity between a reference and test signal was rated for the quality of *coloration* on a scale from 0 to 100, whereby a rating of 100 means entirely the same (no coloration). In this study, perceived coloration is therefore derived as the flipped ratings of perceived similarity

Table 2. Details of the binaural stimuli used in the evaluation.*

Test	Number of stimuli	Number of source positions	Audio content	Coloration type	Number of participants
Mc18 [47]	48	8	Pink noise	Equalized binaural Ambisonic HRIRs	20
Mc19a 1 [48]	42	2	Pink noise	Equalized binaural Ambisonic HRIRs	20
Mc19a 2 [48]	21	**Multiple**	Pink noise bursts and train station recording	Equalized binaural Ambisonic HRIRs	20
Mc19b 1 [49]	15	45 (moving)	Pink noise	Preprocessed binaural Ambisonic HRIRs	20
Mc19b 2 [49]	15	**8**	Percussion recordings	Preprocessed binaural Ambisonic HRIRs	20
Mc22 [13]	56	8	Pink noise	10 band-equalized HRIRs	9
Ll22 1 [50]	25	5	Pink noise	HRIRs made wearing different headphones	15
Ll22 2 [50]	25	5	Speech recordings	HRIRs made wearing different headphones	15
Ll22 3 [50]	5	**45**	Rainfall recordings	HRIRs made wearing different headphones	15

* Bold source position number indicates simultaneous playback of multiple sounds.

(0 indicates entirely the same, and 100 denotes as colored as the low anchor). For each trial, the reference stimulus was a convolution of the monophonic audio content (see again Table 2) with HRTFs, and anchors were a low-pass filtered version of the reference with a cut-off frequency $f_o = 3.5$ kHz. Additionally, Mc18, Mc19a and Mc22 featured a medium anchor: a low-pass–filtered version of the reference with $f_o = 7$ kHz. Test stimuli were presented statically (fixed head orientation) over headphones in quiet listening rooms. All tests used open-back headphones: four out of the five tests (Mc18, Mc19a, Mc19b, and Ll22) used Sennheiser HD 650 headphones, while one used Beyerdynamic DT 990 Pro headphones (Mc22).

To predict the coloration, the stimuli (without headphone equalization) were adjusted to an average ITU-R BS.1770-4 loudness [51] of –20 dBFS separately for each experiment and audio content as listed in Table 2. The loudness normalization was done using MATLAB's *integratedLoudness* function.

3.2 Predicting Binaural Coloration

Linear regressions for each model were performed once using the pooled data containing 252 stimuli and once separately for the stimuli from the five studies listed in Table 2. To ease the comparison of the model outputs, the model predictions were transformed separately for each model as

$$C_p = a\,C + b\,, \tag{20}$$

with C being the raw model output, C_p the predicted coloration, and a and b the slope and intercept from the linear regression listed in Table 3. After this transformation, all models produced values roughly between 0 and 100, whereas the range of output values significantly differed between models before the transformation, which is evident from the range of the slope values in Table 3.

The decision to perform only simple linear regressions as a starting point was made because this is how existing models are used in the vast majority of cases. In addition, all models are physiologically inspired, in which case a linear relationship between model predictions and perception is to be expected. More elaborate possibilities for relating model predictions and perceptual ratings are discussed in SEC. 4 but are beyond the scope of the current study.

Table 3. Regression slope a and intercept b for each model.

Model	a	b
BSD	4.803	13.379
LSD	4.101	10.252
$\mathrm{BSD_A}$	13.029	10.941
$\mathrm{Moore04_{SD}}$	37.200	13.262
$\mathrm{Moore04_{sum}}$	0.673	8.261
CLL	11.240	16.230
ISO 532-1	159.228	6.883
ISO 532-2	199.869	12.151
PBC-1	9.649	5.701
$\mathrm{PBC\text{-}2_{mean}}$	15.766	–1.367
$\mathrm{PBC\text{-}2_{interaural}}$	15.607	–0.646
$\mathrm{PBC\text{-}2_{level}}$	15.192	–0.115

3.3 Results

The regression model results for each tested coloration method are presented for each listening test individually in Table 4 and for all listening tests pooled together in Table 5. Three measures are reported: $R^2_{\mathrm{adj.}}$, the proportion of explained variance (the squared correlation coefficient but corrected for the number of predictors, which is one in this case); $|\bar{E}|$, the mean absolute error between the ratings and the predictions; and $\max|E|$, the maximum absolute error between the ratings and predictions. Higher $R^2_{\mathrm{adj.}}$ values indicate more accurate predictions of the listening test results, and lower $|\bar{E}|$ and $\max|E|$ values indicate better fits.

All models produced positive correlation between predicted and perceived coloration. The $R^2_{\mathrm{adj.}}$ values show that for most of the tested listening tests, all models produce a good result. The Mc18 test, in general, produced the lowest $R^2_{\mathrm{adj.}}$ for all models, although PBC-2 produced the highest $\mathrm{R}^2_{\mathrm{adj.}}$ values. For three out of the five listening tests, the PBC-2 model produced the highest $\mathrm{R}^2_{\mathrm{adj.}}$ values and the lowest $|\bar{E}|$ and $\max|E|$ values. For the two tests where PBC-2

Table 4. Results from the linear regression between the predicted and perceived coloration.*

	Mc18			Mc19a			Mc19b			Mc22			Ll22		
Model	$R^2_{adj.}$	$\lvert\bar{E}\rvert$	max $\lvert E\rvert$	$R^2_{adj.}$	$\lvert\bar{E}\rvert$	max $\lvert E\rvert$	$R^2_{adj.}$	$\lvert\bar{E}\rvert$	max $\lvert E\rvert$	$R^2_{adj.}$	$\lvert\bar{E}\rvert$	max $\lvert E\rvert$	$R^2_{adj.}$	$\lvert\bar{E}\rvert$	max $\lvert E\rvert$
BSD	0.05	6.85	20.7	0.78	7.02	24.4	0.83	3.68	12.3	0.69	9.7	46.9	0.66	11.7	25.8
LSD	0.05	6.87	20.9	0.74	7.69	25.4	0.81	3.91	12.4	0.44	13.7	52.1	0.68	11.3	26.5
BSD_A	0.16	6.63	17.8	0.89	4.85	15.2	0.82	3.86	10.9	0.91	5.17	23.8	0.76	10.1	**23.6**
$Moore04_{SD}$	0.00	7.12	21.5	0.75	7.01	26.3	0.66	5.67	15.8	0.91	5.27	21.1	0.62	12.6	34.0
$Moore04_{sum}$	0.17	6.39	18.3	0.90	4.86	15.0	0.74	4.73	14.7	0.91	4.92	23.4	0.72	11.0	25.2
CLL	0.35	5.95	15.2	0.85	5.94	17.3	0.35	8.11	23.0	0.81	7.74	35.5	0.74	10.3	24.1
ISO 532-1	0.25	6.57	16.5	**0.91**	**4.65**	**14.2**	0.65	5.60	15.5	0.90	5.34	22.0	0.77	9.18	27.2
ISO 532-2	0.35	6.13	**14.1**	0.88	5.54	15.6	0.33	8.52	19.0	0.82	7.92	33.6	0.79	9.05	26.1
PBC-1	0.17	6.50	17.5	0.87	5.31	18.2	0.84	3.80	12.0	0.87	6.09	29.2	0.70	11.2	26.2
PBC-2 mean	0.36	5.90	16.9	0.85	5.67	18.6	0.81	4.20	14.4	0.92	5.22	19.2	0.82	8.25	23.9
PBC-2 interaural	**0.49**	**5.29**	14.4	0.85	5.53	18.6	**0.87**	**3.34**	**9.83**	0.92	5.07	**17.6**	0.86	7.11	25.1
PBC-2 level	0.44	5.53	15.5	0.86	5.53	17.9	0.86	3.48	10.5	**0.92**	**4.91**	18.3	**0.86**	**7.10**	24.9

* $R^2_{adj.}$ denotes the adjusted explained variance, $\lvert\bar{E}\rvert$ denotes mean absolute error, and max $\lvert E\rvert$ denotes maximum absolute error. The highest $R^2_{adj.}$ and lowest $\lvert\bar{E}\rvert$ and max $\lvert E\rvert$ values for each listening test are displayed in bold.

Table 5. Results from the linear regression between the predicted and perceived coloration for all listening test data pooled.*

Model	$R^2_{adj.}$	$\lvert\bar{E}\rvert$	max $\lvert E\rvert$
BSD	0.48	11.7	47.5
LSD	0.44	12.1	48.5
BSD_A	0.72	8.35	30.1
$Moore04_{SD}$	0.42	12.2	44.7
$Moore04_{sum}$	0.64	9.50	38.1
CLL	0.61	9.87	44.1
ISO 532-1	0.78	7.43	28.0
ISO 532-2	0.70	8.64	41.8
PBC-1	0.66	9.35	41.3
PBC-2 mean	0.83	6.45	22.2
PBC-2 interaural	0.85	6.09	**21.9**
PBC-2 level	**0.85**	**6.04**	21.9

* $R^2_{adj.}$ denotes the adjusted explained variance, $\lvert\bar{E}\rvert$ denotes mean absolute error, and max $\lvert E\rvert$ denotes maximum absolute error. The highest $R^2_{adj.}$ and lowest $\lvert\bar{E}\rvert$ and max $\lvert E\rvert$ values are displayed in bold.

did not produce the highest $R^2_{adj.}$ values (Mc19a and Mc22), the explained variance was maximally 5% below that of the best model.

Table 5 shows the data of all five listening tests pooled together, indicating consistency and the overall performance of the models. Here, all three versions of the PBC-2 model perform the best, producing the highest $R^2_{adj.}$ values, around 84%, along with the lowest $\lvert\bar{E}\rvert$ and max $\lvert E\rvert$ values. Interestingly, ISO 532-1 is the second-highest performer, with an explained variance of 78%, and the BSD using an auditory filterbank (BSD_A) is third, with 72% explained variance. The three models with the lowest explained variance, around 45%, are LSD, $Moore04_{SD}$, and BSD.

To further understand the results of the regression analysis and how different listening tests compare, Fig. 2 presents the predicted versus perceived coloration and the prediction error for selected models. It is again clear that all three variants of the proposed PBC-2 model produce the highest correlation between predicted and perceived coloration values. For the other models, certain trends appear. The coloration values for Mc18 tended to be overestimated for almost all tested models. This includes PBC-2 with the mean binaural weighting (simple mean average of the left and right coloration values). This seems improved for the two versions of PBC-2 with dedicated binaural weighting stages, however. In fact, for these PBC-2 versions, no test appears systematically over- or underestimated.

A second trend that can be observed for all models except PBC-2 is the overestimation of stimuli with very low coloration. Before applying Eq. (20), all models output zero in case of identical input signals. The bias for overestimation of low coloration is thus well reflected in the intercepts (b coefficients) listed in Table 3. This is also reflected in the prediction errors, which are not equally distributed around zero in these cases. The effect is most pronounced for CLL, which predicts a minimum coloration of 17.1 (intercept of 16.2) and is discussed in more detail in the next section.

4 DISCUSSION

While all tested coloration models appear to predict binaural coloration relatively accurately, the overall variance that was explained by the model predictions using the pooled data was highest for the proposed PBC-2 methods, and the explained variance from regressions computed separately for stimuli from each listening test was consistently high when using the PBC-2 model, whereas most other models performed poorly on at least one test. This suggests that binaural coloration values predicted using the PBC-2 method are more comparable between tests, which is also illustrated by the closest following of the regression line in Fig. 2. The second-best performing ISO 532-1 model explains 9% less variance and exhibits 6% larger maximum absolute error value and 1% larger mean absolute error (see again Table 5). Statistical tests based on the Pearson correlation coefficients, which are proportional to $R^2_{adj.}$, showed that the PBC-2 performed significantly better than ISO 532-1 ($p < 0.001$ for all tests listed in [52, Table 2]).

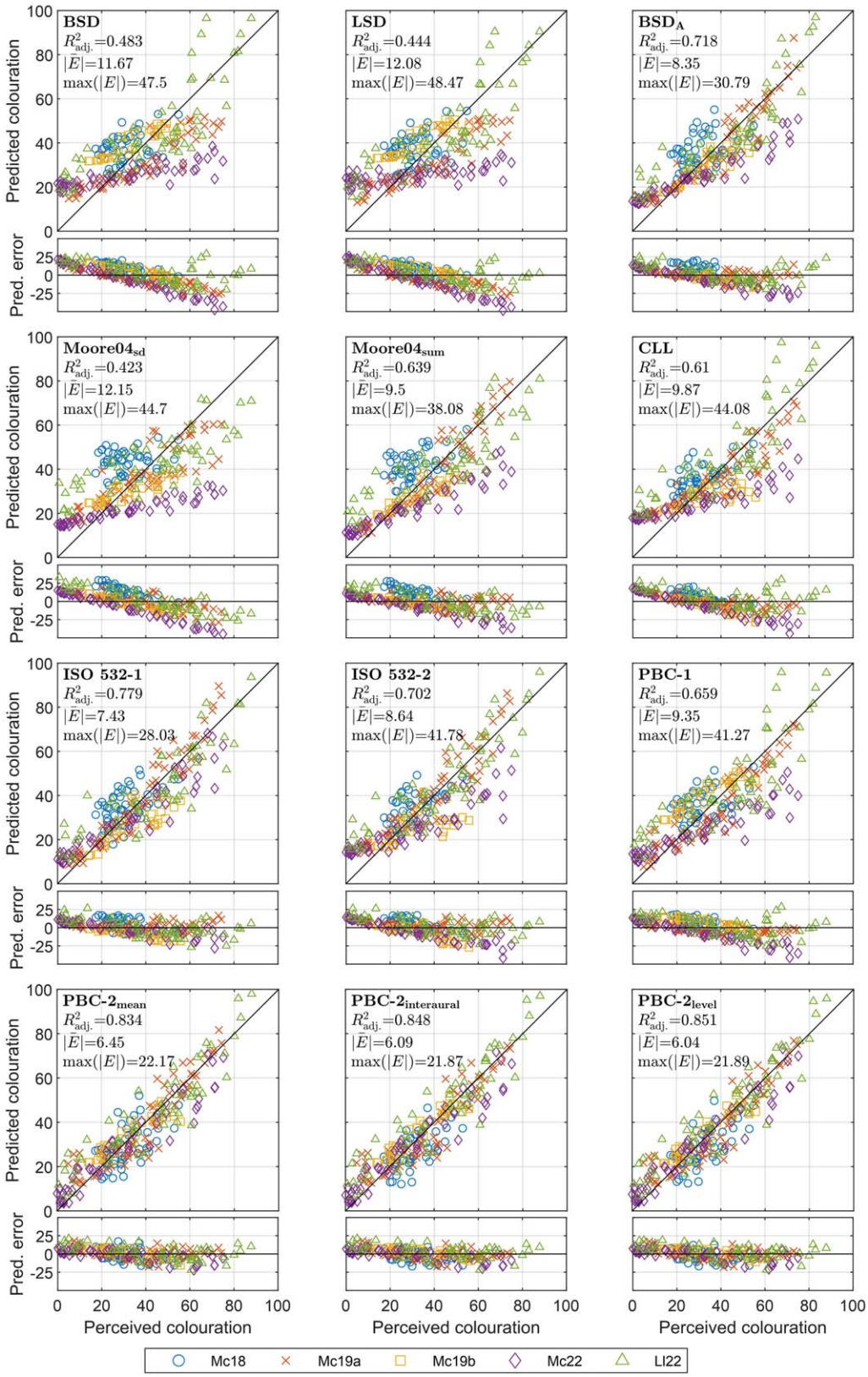

Fig. 2. Model performance by means of the predicted versus rated coloration and the prediction error (pred. error) for each model.

A sensitivity analysis was conducted to gain insights into the PBC-2 model's performance. The full-wave rectification retains all signal information, and using half-wave or no rectification resulted in a 1 and 9% decrease in the explained variance, respectively. Similarly, avoiding the averaging of amplitude values over frequency bands and implementing one-third octave smoothing only above 5 kHz retains more spectral details, especially at low frequencies. Using no spectral smoothing, 8.9% less variance is explained, and applying broadband smoothing. The ERB

frequency-dependent magnitude weighting is more smooth than a 1-kHz low-pass filter, and the additional weighting factor w_{ERB} greater emphasizes low frequencies; setting $w_{\mathrm{ERB}} = 1$ causes a decrease of 7% in the explained variance. In total, 15.3% less variance is explained if omitting the rectification, smoothing, and setting $w_{\mathrm{ERB}} = 1$. In all cases except for using half-wave instead of full-wave rectification, the induced differences were statistically significant, with $p < 0.05$ for all cases and the loss of explained variance exceeding the difference between the PBC-2 and ISO 532-1 models, which is 7%.

Three binaural weighting methods were implemented in the PBC-2 model. The first was a simple mean, the second was based on interaural level and time differences, and the third was based on ILDs computed from the maximum level across reference and test stimuli. The results show that the three methods are fairly comparable overall; however, the two dedicated binaural weighting stages explain up to 20% more variance for perceptual ratings from Mc18 and are statistically significantly better than taking the simple mean ($p < 0.05$ for all cases). As for which of the two dedicated binaural stages is better, no clear judgment can be made yet because differences were not significant, with $p > 0.14$ cases. Similarly, ISO 532-1 also performed well for the pooled data, even though binaural coloration values were calculated from a simple mean of left and right coloration values. One reason for the seemingly low effect of the binaural stages is that the test signals used did not necessarily feature coloration only in one ear—this tends not to occur very often in real-world signals. Future evaluation should investigate this further.

A point to note is that perceived coloration values can vary significantly with listening test paradigm. To produce ratings that can be compared across tests, it is paramount to have consistency in the test paradigm, including potential reference and anchor stimuli. The five listening tests used in this evaluation were all MUSHRA-like tests that used HRTF convolution reference signals and anchor signals that were 3.5-kHz low-passed versions of the reference. This fostered the comparability of ratings across tests and made it possible to validate the models using an extended pool of perceptual ratings. At the same time, the anchor choice most likely decreased content effects that were often observed in other studies. For example, Ben-Hur et al. [53] found significantly lower perceived coloration for speech compared with noise. Future versions of the PBC should account for this, which would possibly require spectrotemporal model stages and listening tests using either no or content-agnostic anchors. It is unclear why the results of Mc18 [47] correlated so poorly with the predictions from the tested models; further work is needed to understand this better.

Although this study used a range of test signals from different listening tests, a greater variety is still needed in future. A total of 186 of the 252 test signals used pink noise. Future evaluation should include more music, both periodic and transient sounds, more speech, and foley sounds such as footsteps, birdsong, and vehicles. A total of 141 of the 252 test signals were binaural Ambisonic renders. A greater variation in types of coloration should be included in the future, such as between HRTFs from different heads and degradations from perceptual coding algorithms (e.g. lossy data compression). A total 211 out of the 252 test signals were a single-source convolved with a single (anechoic) HRTF. Reverberant signals should be considered too, as should more real-world complex scenes with moving sources of varying source width. Finally, the listening tests used in this study primarily featured white, male participants between the ages 20 and 40. This is not representative of the general population and should be addressed, especially if fitting a model's parameters to the listening test results.

In this study, coloration was intentionally predicted using simple linear models because this is how these models are often used. As a consequence, some models could not unfold their full potential and showed systematic biases, such as an overestimation at the low end of the scale, that might be mitigated when using more complex models. Potential solutions to this problem would be nonlinear regression (curve fitting), multiple linear regression, or principle component regression as done by Doma et al. [25] for predicting just noticeable differences, or machine learning–based prediction as suggested by Jiang et al. [23] for predicting the overall audio quality. The latter approach, however, would clearly require a larger stimulus pool than that for which rated coloration is available and would take away the explanatory power of the model.

A significant number of features were implemented and trialed in the development of the PBC-2 model that were ultimately excluded due to their minimal effect on predicted coloration values. The features reported in SEC. 2 had the most significant effect. The most notable exclusion was equal loudness contours (ISO 226), although other exclusions included signal compression, temporal smoothing, and temporal masking. It is possible that, were the variety of test signals to be even more diverse, these features could have more effect. No signals made explicit use of temporal masking, for example. In future testing with more varied stimuli, sounds that specifically target these features of the human auditory system could be included.

5 CONCLUSIONS

This paper has presented an improved PBC-2 auditory model for predicting the coloration between binaural signals, using digital signal processing techniques to model the binaural auditory system. The method has been compared with nine commonly used methods for predicting binaural coloration, using the results and stimuli of five previously conducted listening tests, totaling 252 paired binaural signals and coloration ratings. The results show that the proposed method is most consistent across tests and shows the highest overall explained variance of 85%.

Future work will look to further expand the evaluation of models to predict binaural coloration. A greater variation in binaural signals is still needed, including complex scenes, variation in ipsilateral and contralateral coloration, reverberant scenes, and coloration due to lossy signal com-

pression. More variation in source signals could also be explored, employing more natural sounds such as transient percussion signals, periodical musical sounds, and wide-band and narrow-band sounds. Finally, additional signal processing steps and modeling stages of the human binaural auditory system make possible additions to the coloration prediction model.

A MATLAB implementation of the PBC-2 method is available for download as part of the AMT under the model name *mckenzie2025* (in the mckenzie2025 branch and, from version 1.7 onward, in the main release), along with associated demonstration and analysis scripts.

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THE AUTHORS

Thomas McKenzie

Fabian Brinkmann

Thomas McKenzie is a lecturer in acoustics and architectural acoustics at the Reid School of Music, Edinburgh College of Art, University of Edinburgh, UK. He received a B.Sc. from the University of Leeds, UK, in 2013, before receiving an M.Sc. and then a Ph.D. in music technology at the University of York, UK, in 2015 and 2020, respectively. He then undertook a position as a postdoctoral researcher at Aalto University, Finland. Thomas is a member of the AES, and his research interests include spatial audio and psychoacoustics.

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Fabian Brinkmann received an M.A. degree in communication sciences and technical acoustics in 2011 and a Dr. rer. nat. degree in 2019 from the Technische Universität Berlin, Germany. He focuses on the fields of signal processing and evaluation approaches for spatial audio. Fabian is a member of the AES, German Acoustical Society (DEGA), and the European Acoustics Association (EAA) technical committee for psychological and physiological acoustics.